WORKED SOLUTION

STEP III 1988, Question 11

Mechanics • Ladder Equilibrium with Friction

STEP III 1988, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q11

The key idea

First sort out the geometry: the ground, the ramp and the ladder make a triangle whose angles you already know, and that tells you which way the smooth ramp pushes on the top of the ladder. After that it is the standard ladder recipe: resolve twice, take moments about the foot, and compare the friction needed with μ times the normal reaction. Because the labourer's position enters only through the moment equation, one general formula answers both parts.

Solution

Setting up

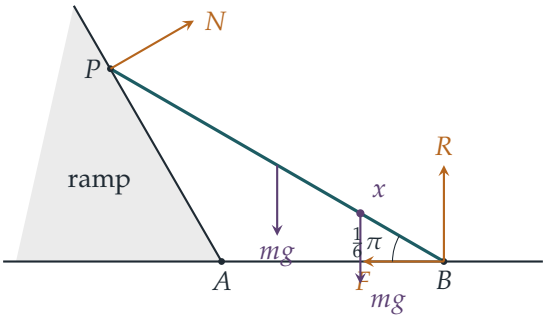
Model the ladder as a uniform rod and the labourer as a particle. Let A be the foot of the ramp, B the foot of the ladder and P its top, on the ramp.

Geometry. In triangle ABP the angle at B is $\frac{1}{6}\pi$ (the ladder's inclination). The ramp makes $\frac{1}{6}\pi$ with the vertical, so it makes $\frac{1}{3}\pi$ with the horizontal on the far side of A. The angle at A inside the triangle is therefore $\pi - \frac{1}{3}\pi = \frac{2}{3}\pi$, which leaves $\frac{1}{6}\pi$ at P. So the triangle is isosceles with $AP = AB$, and the sine rule gives

$$AB = \frac{l \sin \frac{1}{6}\pi}{\sin \frac{2}{3}\pi} = \frac{l}{\sqrt{3}},$$

which matches the given distance. The data are consistent, and P really is on the ramp.

Direction of the ramp's reaction. The ramp is smooth, so its reaction N on the ladder is perpendicular to the ramp. The ramp is at $\frac{1}{3}\pi$ to the horizontal, so its normal is at $\frac{1}{6}\pi$ to the horizontal, pointing up and away from the ramp, towards B. So N has horizontal component $\frac{\sqrt{3}}{2}N$ pushing the ladder away from the ramp, and vertical component $\frac{1}{2}N$ upwards.



Equations. Let the labourer be a distance x up the ladder from B. At B the ground gives normal reaction R upwards and friction F horizontally towards the ramp (the only horizontal force otherwise is the component of N, which pushes the foot away from the ramp, so friction must oppose that).

Resolving horizontally and vertically:

$$F = \tfrac{\sqrt{3}}{2}N, \qquad\qquad R + \tfrac{1}{2}N = 2mg.$$

Taking moments about B, so that R and F drop out. The horizontal distance from B to a point a distance s up the ladder is $s\cos\frac{1}{6}\pi = \frac{\sqrt{3}}{2}s$. The line of action of N is at $\frac{1}{6}\pi$ to the horizontal, while BP is at $\frac{1}{6}\pi$ on the other side of the horizontal, so N makes an angle $\frac{1}{3}\pi$ with the ladder. Its moment about B is $Nl\sin\frac{1}{3}\pi = \frac{\sqrt{3}}{2}Nl$. Hence

$$\tfrac{\sqrt{3}}{2}Nl = mg\cdot\tfrac{\sqrt{3}}{2}\cdot\tfrac{l}{2} + mg\cdot\tfrac{\sqrt{3}}{2}x \quad\Longrightarrow\quad N = mg\left(\frac{1}{2}+\frac{x}{l}\right).$$

Ask yourself Why does the moment of N come out so simply?

The perpendicular distance from B to the line of action of N is $l\sin\frac{1}{3}\pi$, because N meets the ladder at $\frac{1}{3}\pi$. Or split N into components: the horizontal part $\frac{\sqrt{3}}{2}N$ acts at height $\frac{1}{2}l$ and the vertical part $\frac{1}{2}N$ acts at horizontal distance $\frac{\sqrt{3}}{2}l$. Both give a moment of $\frac{\sqrt{3}}{4}Nl$, and these add to $\frac{\sqrt{3}}{2}Nl$.

Then $R = 2mg - \frac{1}{2}N = mg\left(\frac{7}{4} - \frac{x}{2l}\right) > 0$ for $0 \le x \le l$, so the foot stays on the ground. The ladder does not slip as long as $F \le \mu R$, which means

$$\mu \;\ge\; \frac{F}{R} = \frac{\sqrt{3}\,N}{4mg - N} = \frac{\sqrt{3}\,(l+2x)}{7l-2x}.$$

This ratio increases with x: the numerator increases and the positive denominator decreases. So the friction needed grows steadily as the labourer climbs.

Part (i)

The ladder slips as soon as the labourer reaches the midpoint. So limiting friction is reached exactly when $x = \frac{1}{2}l$, and since F/R increases with x, not before. Then $N = mg$,

$F = \frac{\sqrt{3}}{2}mg$ and $R = \frac{3}{2}mg$, so

$$\mu = \frac{F}{R} = \frac{\sqrt{3}/2}{3/2} = \frac{1}{\sqrt{3}}.$$

Part (ii)

The labourer reaches the top if $F \leq \mu R$ for every $x \in [0, l]$. Since F/R increases with x, the worst case is $x = l$. There $N = \frac{3}{2}mg$, $F = \frac{3\sqrt{3}}{4}mg$ and $R = \frac{5}{4}mg$, so the minimum coefficient is

$$\mu = \frac{3\sqrt{3}/4}{5/4} = \frac{3\sqrt{3}}{5} \approx 1.04.$$

Checks

- With nobody on the ladder ($x = 0$) the formula gives $\mu \geq \sqrt{3}/7 \approx 0.247$. That is smaller than both answers, as it should be.
- Substituting $l = 1$, $m = 1$, $g = 9.8$ into the three equilibrium equations numerically, with the forces written as vectors, reproduces $N/mg = 1$ and $\frac{3}{2}$ and the ratios $F/R = 0.57735\ldots$ and $1.03923\ldots$. It also confirms that F/R increases at 101 positions up the ladder.

Common slips

- Taking the ramp's reaction to be horizontal, as for a vertical wall. Here it is tilted $\frac{1}{6}\pi$ above the horizontal, and its upward component reduces R.
- Forgetting the labourer's weight when resolving vertically, so that $R + \frac{1}{2}N = mg$ instead of $2mg$.

Beyond STEP

Part (ii) is a small optimisation over a configuration: you need the worst case of F/R over all positions of the load. This is how engineers find the critical load case for a structure. A useful way to picture the condition $F \leq \mu R$ is the *cone of friction*: the total ground reaction must lie within an angle $\arctan \mu$ of the normal, and the ladder slips at the moment the required reaction leaves that cone. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction carefully.

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